

Homework 3.1

Problem 2

Consider the bisection method starting with the interval $[1.5, 2.5]$.

a) What is the width of the interval at the n th step?

At $n = 0$, our interval is going to be $[1.5, 2.5]$. At $n = 1$, we will have that the interval will become either $[1.5, c]$ or $[c, 2.5]$ where c is the midpoint of this interval. Thus, at this step, the length of the interval is half of the original. By induction, we can get that the interval length at the n th step will be $\frac{1}{2^n} * |I_0|$ or $\frac{1}{2^n}$ for this starting interval since $|I_0| = |2.5 - 1.5| = 1$.

b) What is the maximum distance between the root r and the midpoint of this interval?

Since we know that a root must exist in the interval, then we have that the farthest a root can be from the center has to be near the boundary, namely close to either a_n or b_n thus we have that the distance has to be either $|c_n - a_n|$ or $|c_n - b_n|$ which both would be half of the interval having length $\frac{1}{2^{n+1}}$.

Problem 4

Determine the formula

$$n \geq \frac{\log(b_0 - a_0) - \log(\epsilon)}{\log(2)} - 1$$

Involving $b_0 - a_0$, and ϵ for the number of steps that must be taken in the bisection method to guarantee that $|r - c_n| \leq \epsilon$

$$|r - c_n| \leq \epsilon \Rightarrow 2^{-(n+1)}(b_0 - a_0) \leq \epsilon$$

$$\log\left(2^{-(n+1)}(b_0 - a_0)\right) \leq \log(\epsilon) \Rightarrow \log(2^{-(n+1)}) + \log(b_0 - a_0) \leq \log(\epsilon)$$

$$\Rightarrow (-n - 1) \log(2) + \log(b_0 - a_0) \leq \log(\epsilon) \Rightarrow -n \log(2) \leq \log(\epsilon) - \log(b_0 - a_0) + \log(2)$$

$$\Rightarrow n \log(2) \geq \log(b_0 - a_0) - \log(\epsilon) - \log(2)$$

$$\therefore n \geq \frac{\log(b_0 - a_0) - \log(\epsilon)}{\log(2)} - 1$$

Problem 11

Give an example (or prove that none exists) in which

$$a_0 < a_1 < a_2 < \dots$$

Since we know that each iteration we half the interval and that $c_n = a_{n+1}$ since we have that $a_n < a_{n+1}$ and $c_n > a_n$, then we can see that the difference between a_n and b_0 is:

$$|a_0 - b_0| = I = \frac{I}{2^0}$$

$$|a_1 - b_0| = \frac{I}{2} = \frac{I}{2^1}$$

...

$$|a_n - b_0| = \frac{I}{2^n}$$

Since $\lim_{n \rightarrow \infty} |a_n - b_0| = 0$, we have that a_n will converge to b_0 thus we cannot have that $a_0 < a_1 < \dots$ rather we will have that $a_0 < a_1 < \dots \leq b_0$.

Problem 12

Give an example in which $a_0 = a_1 < a_2 = a_3 < \dots$

Start with the interval $[0,1]$,

$$a_0 = 0, b_0 = 1$$

$$a_1 = 0, b_1 = \frac{1}{2}$$

$$a_2 = \frac{1}{4}, b_2 = \frac{1}{2}$$

$$a_3 = \frac{1}{4}, b_3 = \frac{3}{8}$$

$$a_4 = \frac{5}{16}, b_4 = \frac{3}{8}$$

$$a_5 = \frac{5}{16}, b_5 = \frac{11}{32}$$

$$a_6 = \frac{21}{64}, b_6 = \frac{11}{32}$$

$$a_7 = \frac{21}{64}, b_7 = \frac{43}{128}$$

So, we see that we are tending to $\frac{1}{3}$, thus taking the function $3x = 1$ will give us our desired sequence.

Computer Problems:

Problem 1

Write and test a subprogram or procedure to implement the bisection algorithm.

```
function Bisection = B(fun,a,b) %This will allow us to use any function
%That we wish to pass through this function.

%This value will be what i will use as my tolerance from the true solution
%f(x) = 0.
delta = 10^(-8);

M = 10^3; %Maximum number of times that this loop can iterate.

for k = 1:M %Iterate up to M times.

    c = (b + a)/2; %Midpoint of the two values.

    if abs(fun(c)) < delta %If our c is good enough, then end program.
        c
        return %end function here if needed.
    end

    if sign(fun(a)) == -sign(fun(c)) %Check the signs and reassing a and b.
        b = c;
    else
        a = c;
    end
end
c %If the loop terminates at M, display the last value it had.
end
```

Test the program on these functions and intervals:

a) $x^{-1} - \tan(x), \left[0, \frac{\pi}{2}\right]$

```
>> format long %So that we can see more digits of precision.

>> ep = 10^(-8); %Since the function is not defined at 0 or at pi/2, we will move a little bit from these
>> %points so that we can do the calculations.

>> f = @(x) (1/x) - tan(x); %We will define the function as such so that we can pass it in our program.

>> B(f,0 + ep,pi/2 - ep) %We have pertubate our interval a bit so that our function is defined.

c = 0.860333589724671
```

```
>> f(0.860333589724671)
```

```
ans = -2.611035831989739e-09
```

```
>> %We can see that this number is within our tolerance of the root.
```

b) $x^{-1} - 2^x, [0,1]$

```
>> g = @(x) (1/x) - 2^x; %We will define the function as such so that we can pass it in our program.
```

```
>> B(g,0 + ep,1) %We will have to perturbate the starting point.
```

```
c = 0.641185741734448
```

```
>> g(0.641185741734448)
```

```
ans = 9.734075989697999e-09
```

```
>> %We can see that this number is within our tolerance of the root.
```